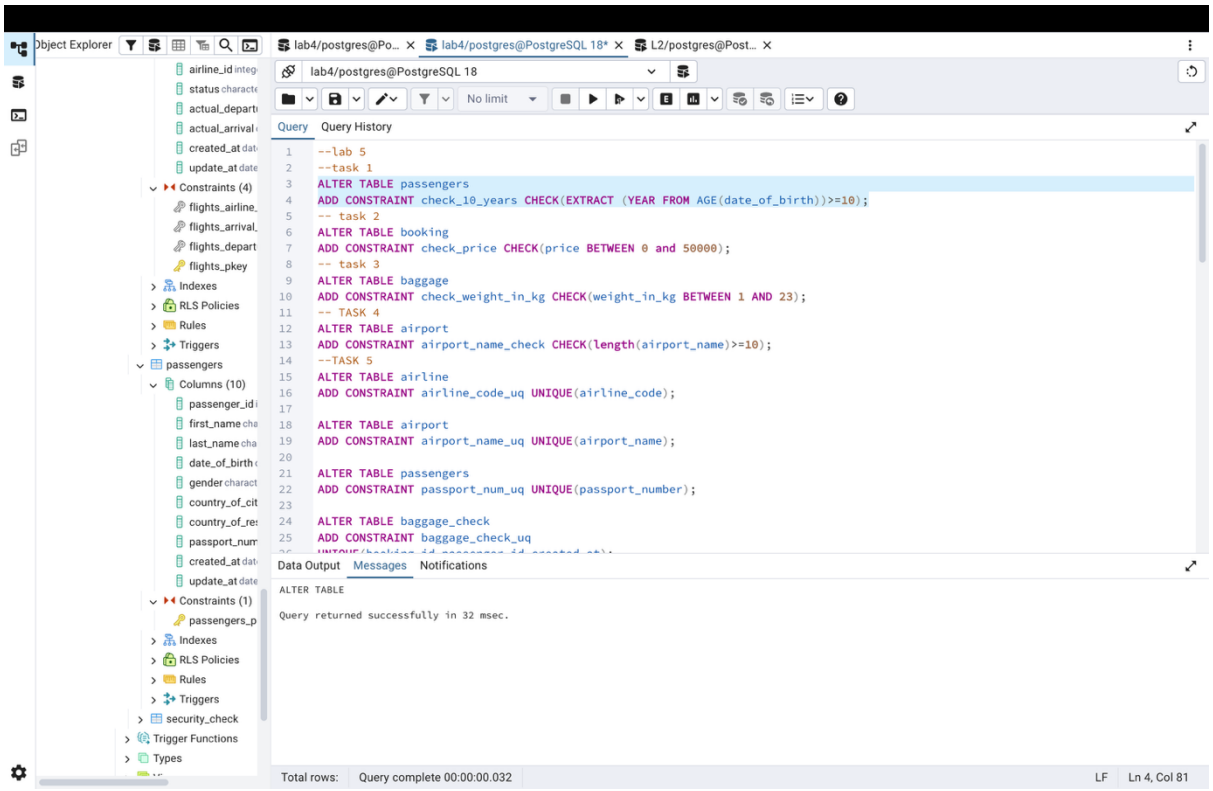
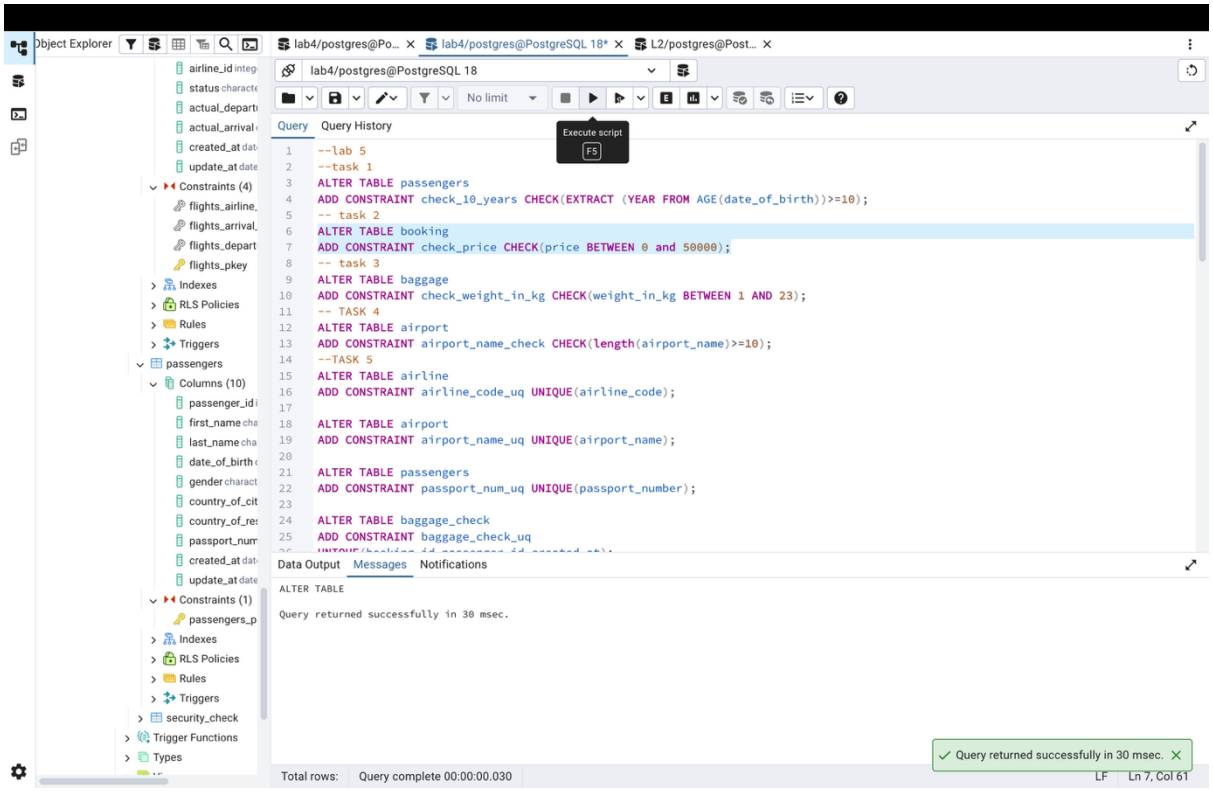
Laboratory Work 5

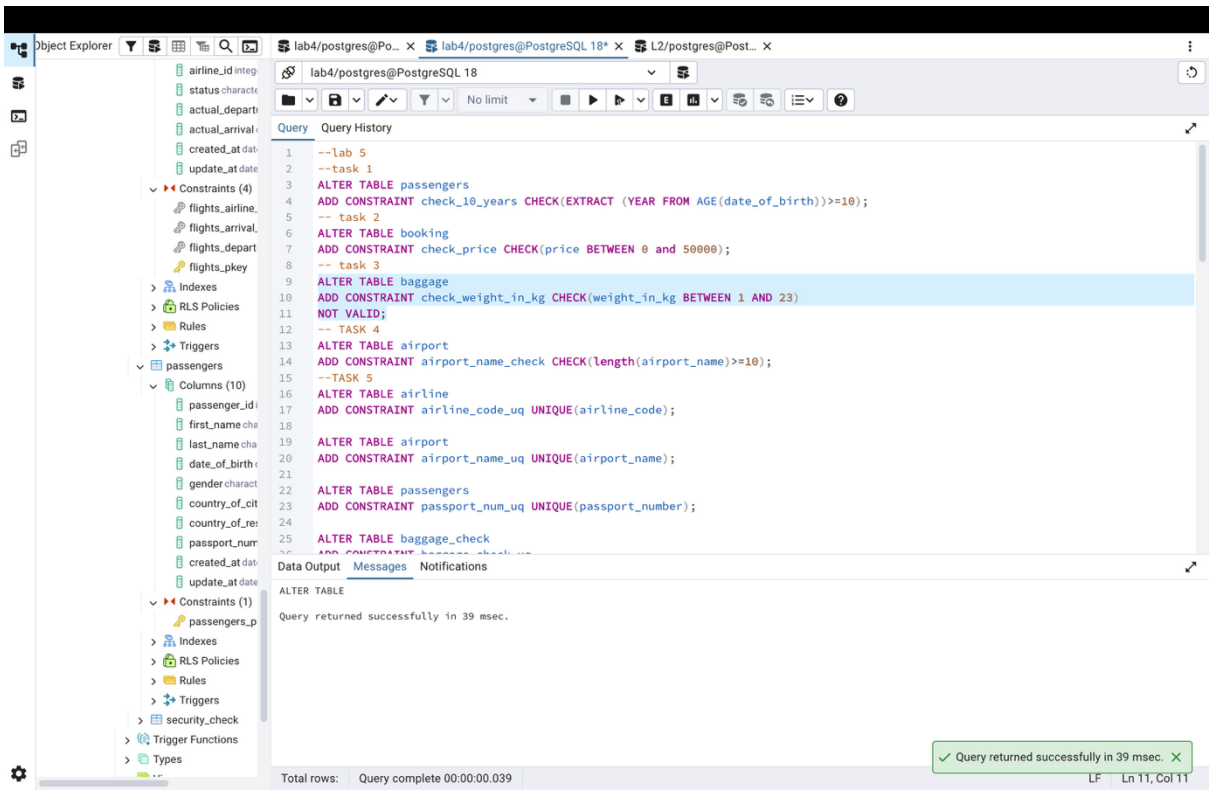
Student: Galym Guliya



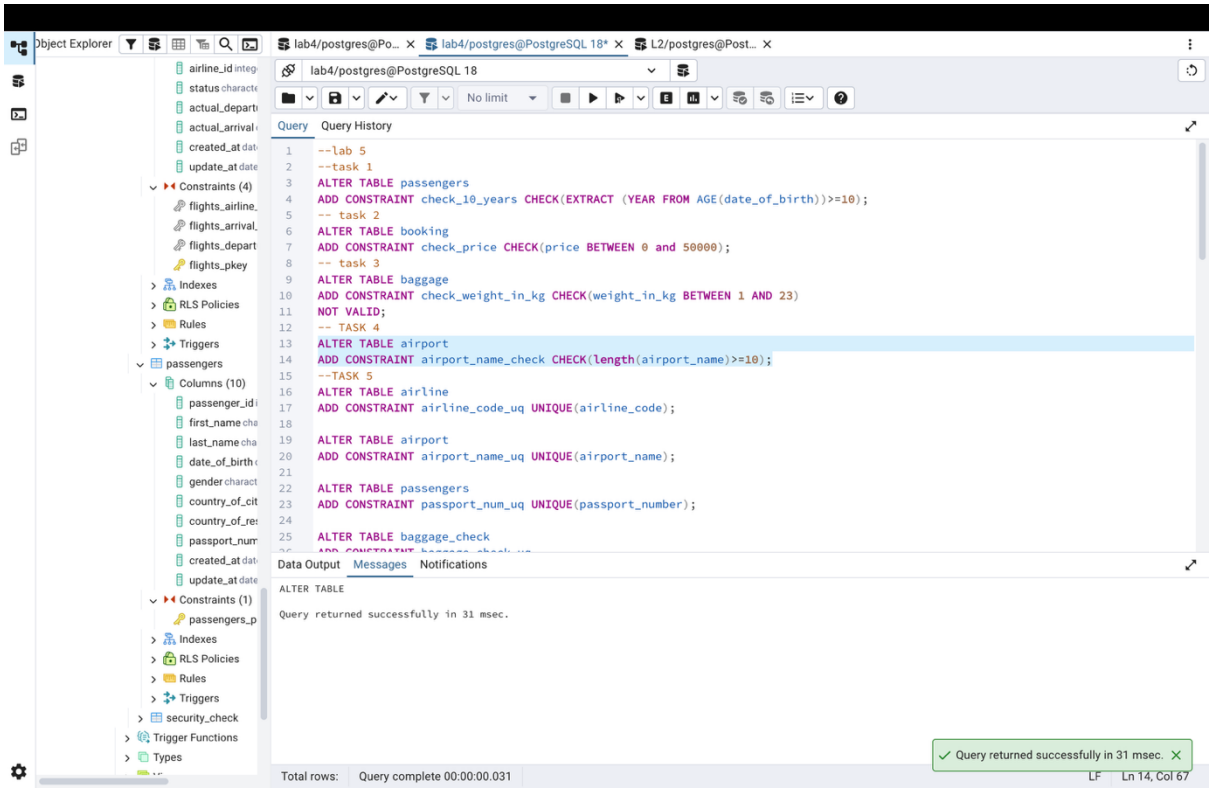
TASK 2


lab4/postgres@PostgreSQL 18*
Query
Query History
--lab 5
--task 1
ALTER TABLE passengers
ADD CONSTRAINT check_10_years CHECK(EXTRACT (YEAR FROM AGE(date_of_birth))>=10);
-- task 2
ALTER TABLE booking
ADD CONSTRAINT check_price CHECK(price BETWEEN 0 and 50000);
-- task 3
ALTER TABLE baggage
ADD CONSTRAINT check_weight_in_kg CHECK(weight_in_kg BETWEEN 1 AND 23);
-- TASK 4
ALTER TABLE airport
ADD CONSTRAINT airport_name_check CHECK(length(airport_name)>=10);
--TASK 5
ALTER TABLE airline
ADD CONSTRAINT airline_code_uq UNIQUE(airline_code);
ALTER TABLE airport
ADD CONSTRAINT airport_name_uq UNIQUE(airport_name);
ALTER TABLE passengers
ADD CONSTRAINT passport_num_uq UNIQUE(passport_number);
ALTER TABLE baggage_check
ADD CONSTRAINT baggage_check_uq
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 30 msec.
Total rows:
Query complete 00:00:00.030
Ln 7, Col 61

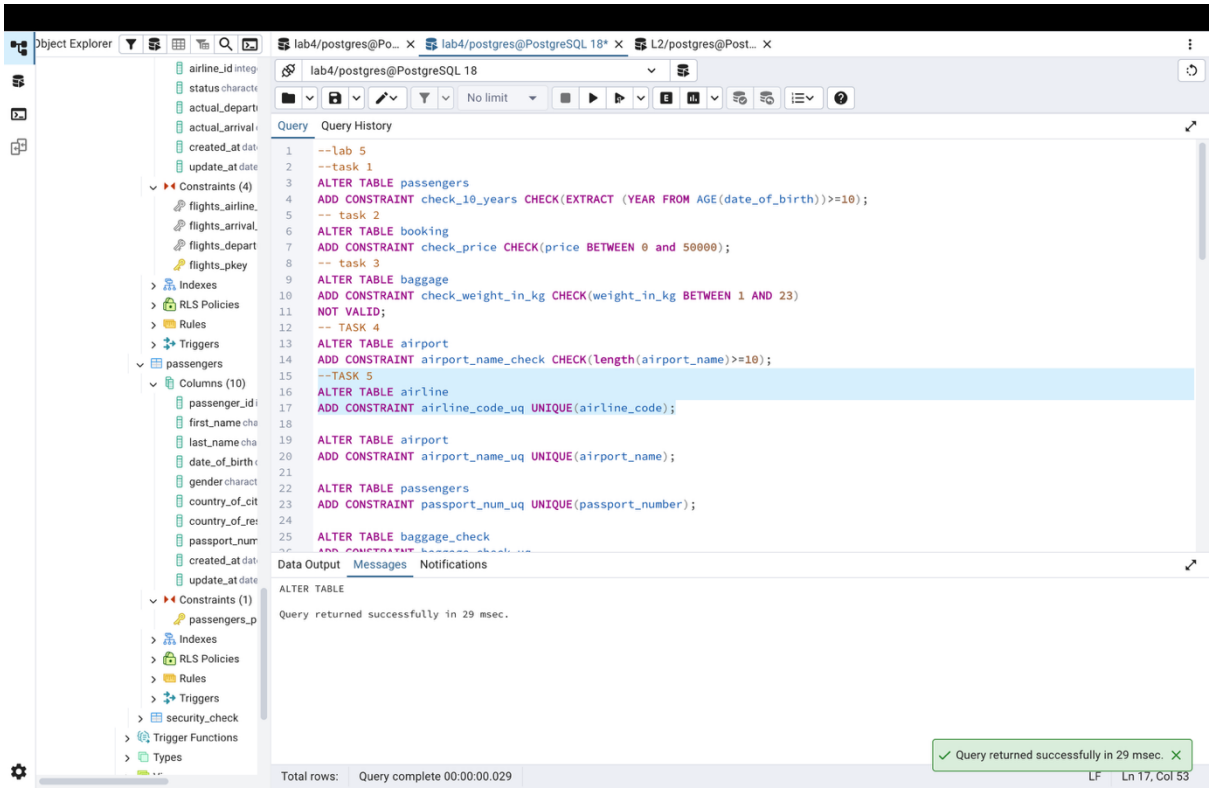
TASK3

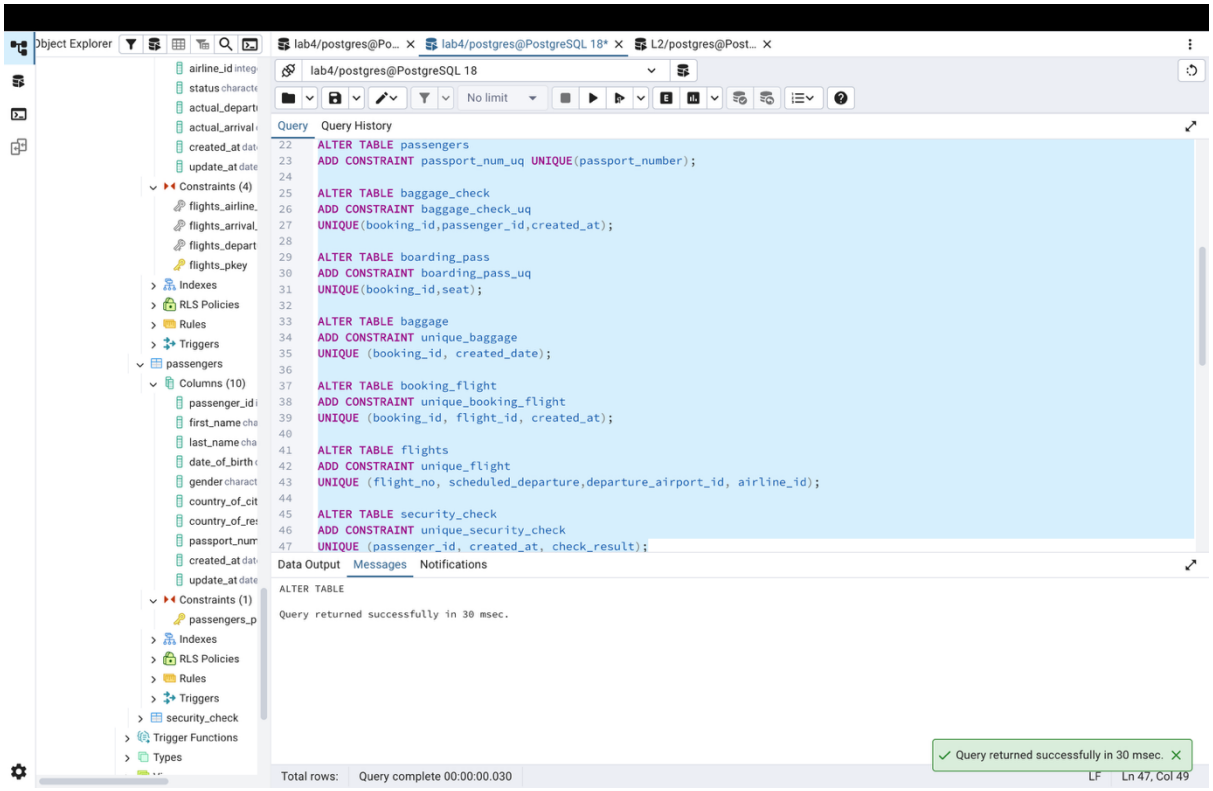

lab4/postgres@PostgreSQL 18*
Query
Query History
--lab 5
--task 1
ALTER TABLE passengers
ADD CONSTRAINT check_10_years CHECK(EXTRACT (YEAR FROM AGE(date_of_birth))>=10);
-- task 2
ALTER TABLE booking
ADD CONSTRAINT check_price CHECK(price BETWEEN 0 and 50000);
-- task 3
ALTER TABLE baggage
ADD CONSTRAINT check_weight_in_kg CHECK(weight_in_kg BETWEEN 1 AND 23)
NOT VALID;
-- TASK 4
ALTER TABLE airport
ADD CONSTRAINT airport_name_check CHECK(length(airport_name)>=10);
--TASK 5
ALTER TABLE airline
ADD CONSTRAINT airline_code_uq UNIQUE(airline_code);
ALTER TABLE airport
ADD CONSTRAINT airport_name_uq UNIQUE(airport_name);
ALTER TABLE passengers
ADD CONSTRAINT passport_num_uq UNIQUE(passport_number);
ALTER TABLE baggage_check
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 39 msec.
Total rows:
Query complete 00:00:00.039
Ln 11, Col 11

TASK 4

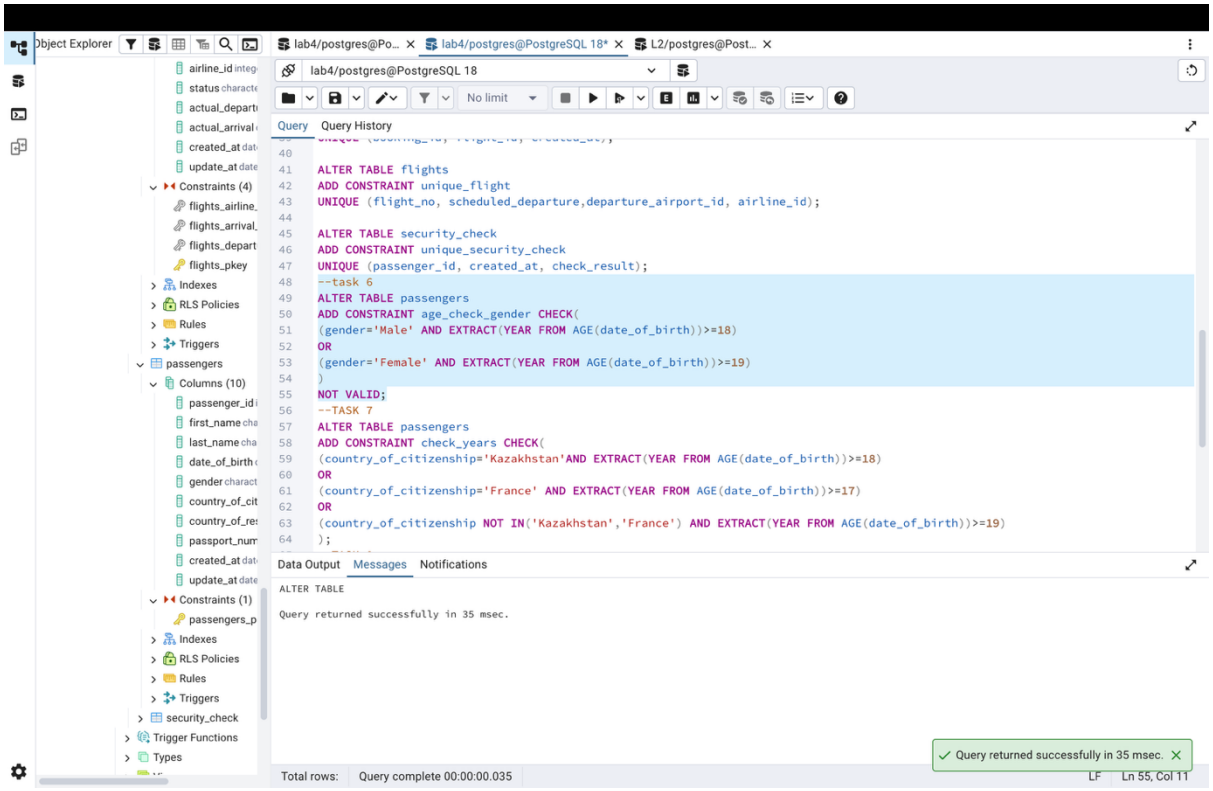

--lab 5
--task 1
ALTER TABLE passengers
ADD CONSTRAINT check_10_years CHECK(EXTRACT (YEAR FROM AGE(date_of_birth))>=10);
-- task 2
ALTER TABLE booking
ADD CONSTRAINT check_price CHECK(price BETWEEN 0 and 50000);
-- task 3
ALTER TABLE baggage
ADD CONSTRAINT check_weight_in_kg CHECK(weight_in_kg BETWEEN 1 AND 23)
NOT VALID;
-- TASK 4
ALTER TABLE airport
ADD CONSTRAINT airport_name_check CHECK(length(airport_name)>=10);
--TASK 5
ALTER TABLE airline
ADD CONSTRAINT airline_code_uq UNIQUE(airline_code);
ALTER TABLE airport
ADD CONSTRAINT airport_name_uq UNIQUE(airport_name);
ALTER TABLE passengers
ADD CONSTRAINT passport_num_uq UNIQUE(passport_number);
ALTER TABLE baggage_check
ALTER TABLE
Query returned successfully in 31 msec.
Query returned successfully in 31 msec.

TASK 5

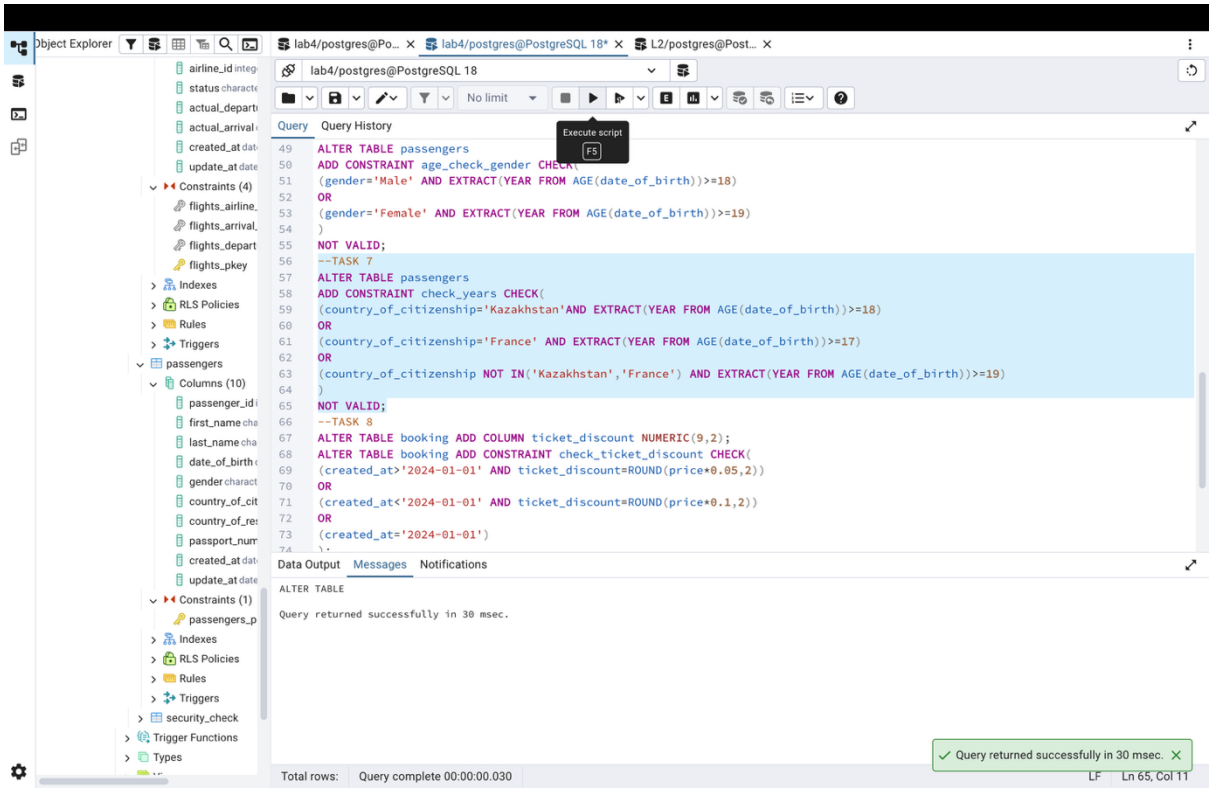

--lab 5
--task 1
ALTER TABLE passengers
ADD CONSTRAINT check_10_years CHECK(EXTRACT (YEAR FROM AGE(date_of_birth))>=10);
-- task 2
ALTER TABLE booking
ADD CONSTRAINT check_price CHECK(price BETWEEN 0 and 50000);
-- task 3
ALTER TABLE baggage
ADD CONSTRAINT check_weight_in_kg CHECK(weight_in_kg BETWEEN 1 AND 23)
NOT VALID;
-- TASK 4
ALTER TABLE airport
ADD CONSTRAINT airport_name_check CHECK(length(airport_name)>=10);
--TASK 5
ALTER TABLE airline
ADD CONSTRAINT airline_code_uq UNIQUE(airline_code);
ALTER TABLE airport
ADD CONSTRAINT airport_name_uq UNIQUE(airport_name);
ALTER TABLE passengers
ADD CONSTRAINT passport_num_uq UNIQUE(passport_number);
ALTER TABLE baggage_check
ALTER TABLE
Query returned successfully in 29 msec.
Query returned successfully in 29 msec.


ALTER TABLE passengers
ADD CONSTRAINT passport_num_uq UNIQUE(passport_number);
ALTER TABLE baggage_check
ADD CONSTRAINT baggage_check_uq
UNIQUE(booking_id,passenger_id,created_at);
ALTER TABLE boarding_pass
ADD CONSTRAINT boarding_pass_uq
UNIQUE(booking_id,seat);
ALTER TABLE baggage
ADD CONSTRAINT unique_baggage
UNIQUE (booking_id, created_date);
ALTER TABLE booking_flight
ADD CONSTRAINT unique_booking_flight
UNIQUE (booking_id, flight_id, created_at);
ALTER TABLE flights
ADD CONSTRAINT unique_flight
UNIQUE (flight_no, scheduled_departure,departure_airport_id, airline_id);
ALTER TABLE security_check
ADD CONSTRAINT unique_security_check
UNIQUE (passenger_id, created_at, check_result);
ALTER TABLE
Query returned successfully in 30 msec.
Query returned successfully in 30 msec.

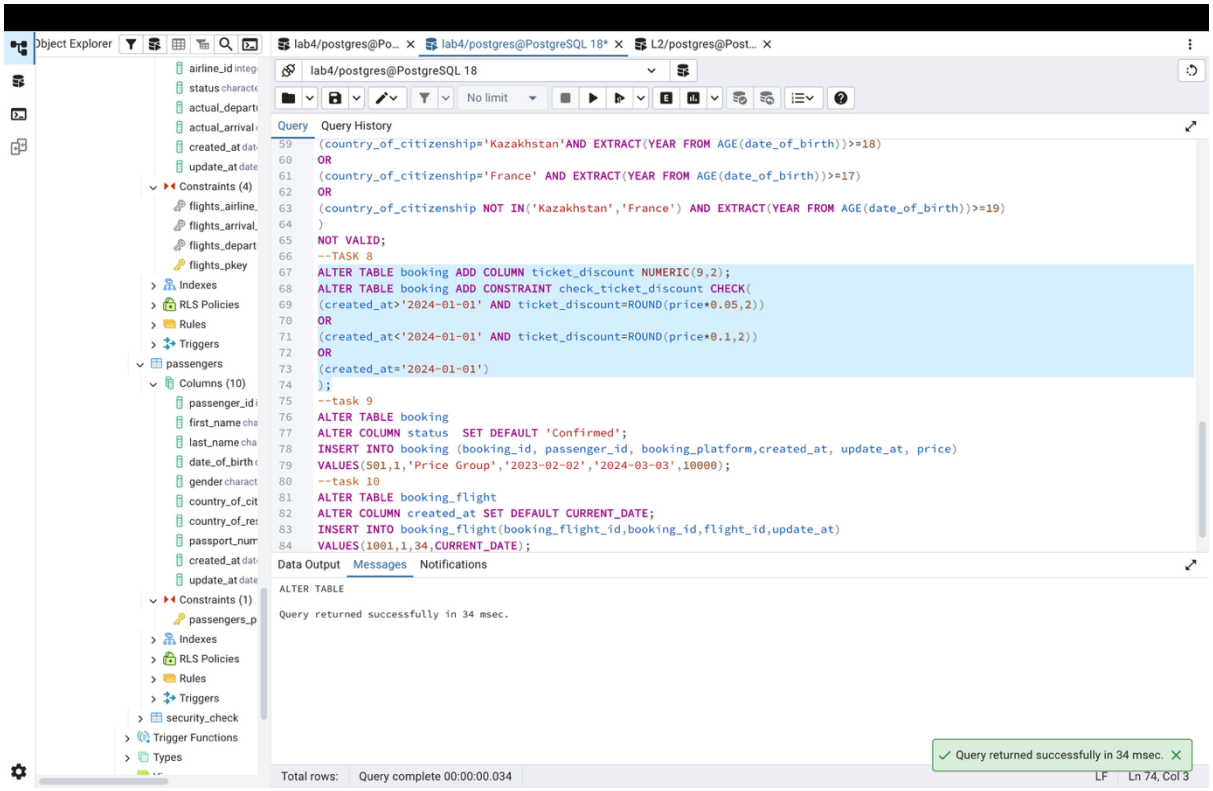
TASK 6


ALTER TABLE flights
ADD CONSTRAINT unique_flight
UNIQUE (flight_no, scheduled_departure,departure_airport_id, airline_id);
ALTER TABLE security_check
ADD CONSTRAINT unique_security_check
UNIQUE (passenger_id, created_at, check_result);
--task 6
ALTER TABLE passengers
ADD CONSTRAINT age_check_gender CHECK(
(gender='Male' AND EXTRACT(YEAR FROM AGE(date_of_birth))>=18)
OR
(gender='Female' AND EXTRACT(YEAR FROM AGE(date_of_birth))>=19)
)
NOT VALID;
--TASK 7
ALTER TABLE passengers
ADD CONSTRAINT check_years CHECK(
(country_of_citizenship='Kazakhstan'AND EXTRACT(YEAR FROM AGE(date_of_birth))>=18)
OR
(country_of_citizenship='France' AND EXTRACT(YEAR FROM AGE(date_of_birth))>=17)
OR
(country_of_citizenship NOT IN('Kazakhstan','France') AND EXTRACT(YEAR FROM AGE(date_of_birth))>=19)
);
ALTER TABLE
Query returned successfully in 35 msec.
Query returned successfully in 35 msec.

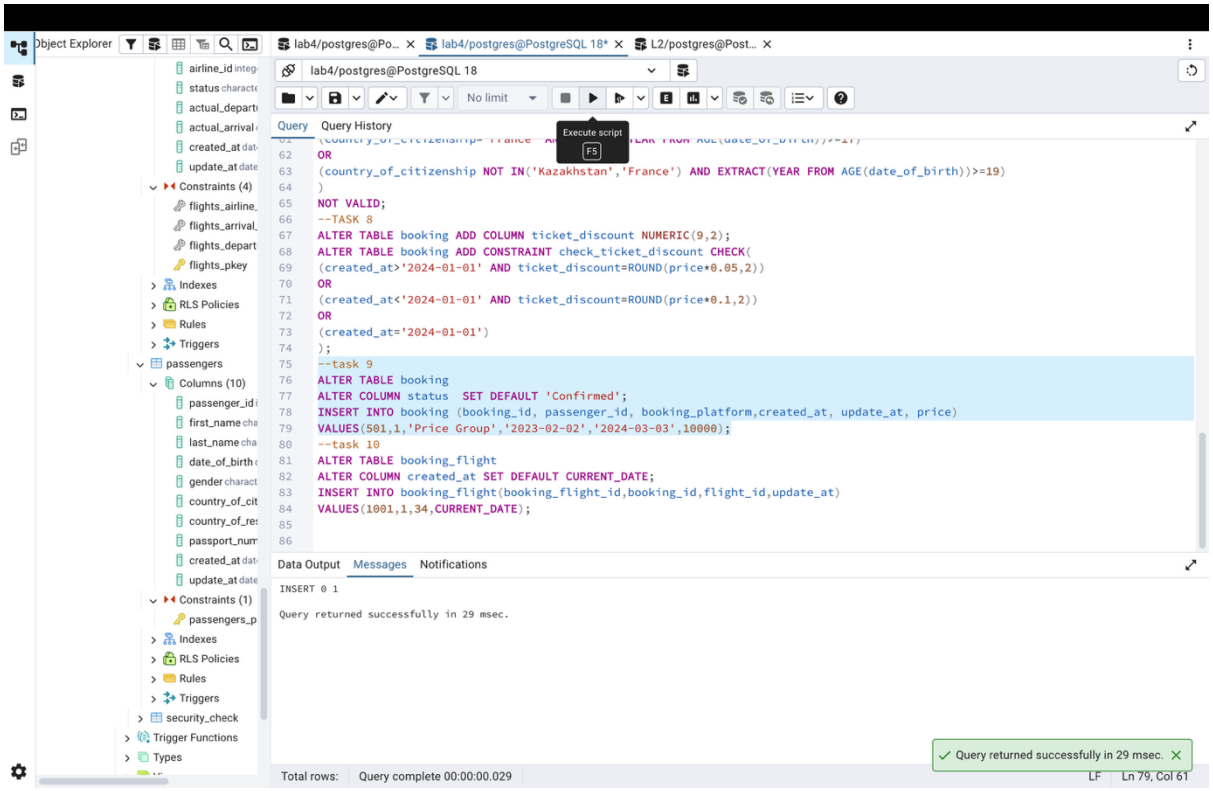
TASK 7


Object Explorer
lab4/postgres@Po...
lab4/postgres@PostgreSQL 18*
L2/postgres@Post...
lab4/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
49 ALTER TABLE passengers
50 ADD CONSTRAINT age_check_gender CHECK(
51 (gender='Male' AND EXTRACT(YEAR FROM AGE(date_of_birth))>=18)
52 OR
53 (gender='Female' AND EXTRACT(YEAR FROM AGE(date_of_birth))>=19)
54)
55 NOT VALID;
56 --TASK 7
57 ALTER TABLE passengers
58 ADD CONSTRAINT check_years CHECK(
59 (country_of_citizenship='Kazakhstan'AND EXTRACT(YEAR FROM AGE(date_of_birth))>=18)
60 OR
61 (country_of_citizenship='France' AND EXTRACT(YEAR FROM AGE(date_of_birth))>=17)
62 OR
63 (country_of_citizenship NOT IN('Kazakhstan','France') AND EXTRACT(YEAR FROM AGE(date_of_birth))>=19)
64)
65 NOT VALID;
66 --TASK 8
67 ALTER TABLE booking ADD COLUMN ticket_discount NUMERIC(9,2);
68 ALTER TABLE booking ADD CONSTRAINT check_ticket_discount CHECK(
69 (created_at>'2024-01-01' AND ticket_discount=ROUND(price*0.05,2))
70 OR
71 (created_at<'2024-01-01' AND ticket_discount=ROUND(price*0.1,2))
72 OR
73 (created_at='2024-01-01')
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 30 msec.
Query returned successfully in 30 msec.
Total rows:
Query complete 00:00:00.030
LF
Ln 65, Col 11

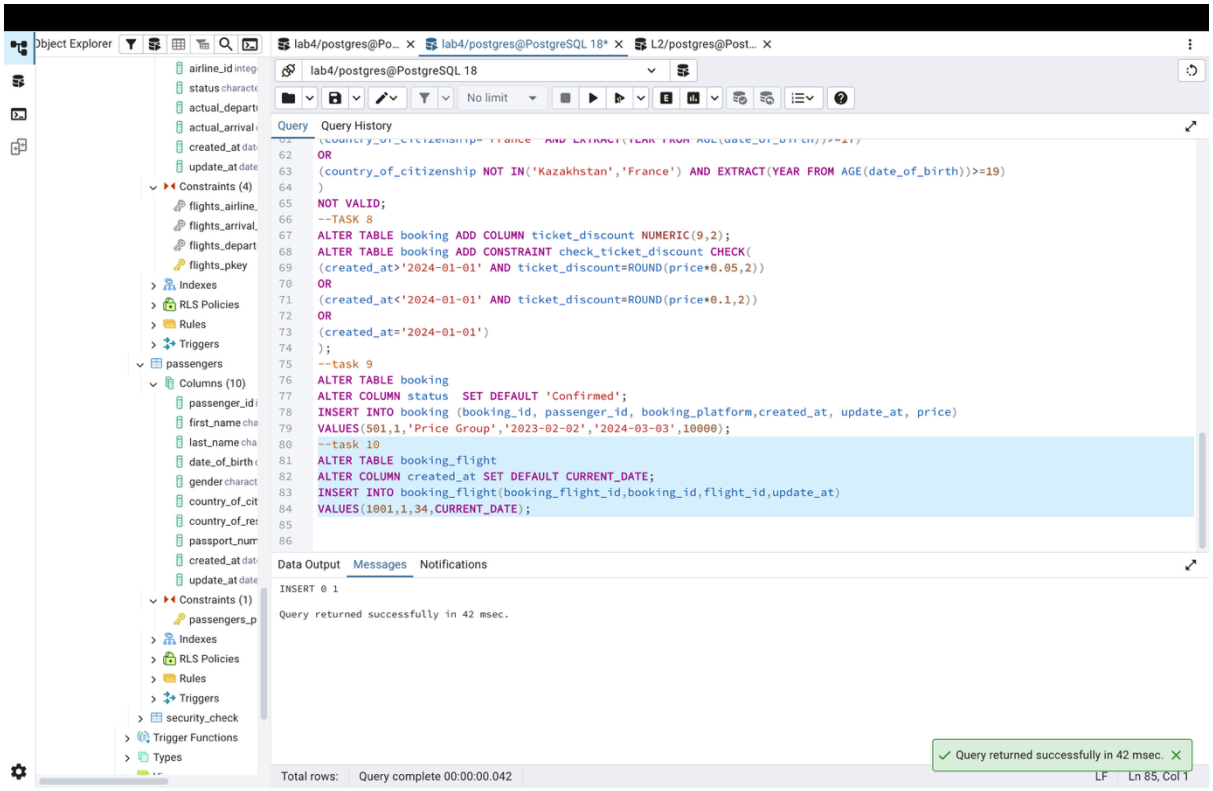
TASK 8


Object Explorer
lab4/postgres@Po...
lab4/postgres@PostgreSQL 18*
L2/postgres@Post...
lab4/postgres@PostgreSQL 18
No limit
Query
Query History
59 (country_of_citizenship='Kazakhstan'AND EXTRACT(YEAR FROM AGE(date_of_birth))>=18)
60 OR
61 (country_of_citizenship='France' AND EXTRACT(YEAR FROM AGE(date_of_birth))>=17)
62 OR
63 (country_of_citizenship NOT IN('Kazakhstan','France') AND EXTRACT(YEAR FROM AGE(date_of_birth))>=19)
64)
65 NOT VALID;
66 --TASK 8
67 ALTER TABLE booking ADD COLUMN ticket_discount NUMERIC(9,2);
68 ALTER TABLE booking ADD CONSTRAINT check_ticket_discount CHECK(
69 (created_at>'2024-01-01' AND ticket_discount=ROUND(price*0.05,2))
70 OR
71 (created_at<'2024-01-01' AND ticket_discount=ROUND(price*0.1,2))
72 OR
73 (created_at='2024-01-01')
74);
75 --task 9
76 ALTER TABLE booking
77 ALTER COLUMN status SET DEFAULT 'Confirmed';
78 INSERT INTO booking (booking_id, passenger_id, booking_platform,created_at, update_at, price)
79 VALUES(501,1,'Price Group','2023-02-02','2024-03-03',10000);
80 --task 10
81 ALTER TABLE booking_flight
82 ALTER COLUMN created_at SET DEFAULT CURRENT_DATE;
83 INSERT INTO booking_flight(booking_flight_id,booking_id,flight_id,update_at)
84 VALUES(1001,1,34,CURRENT_DATE);
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 34 msec.
Query returned successfully in 34 msec.
Total rows:
Query complete 00:00:00.034
LF
Ln 74, Col 3

TASK 9


lab4/postgres@PostgreSQL 18*
Execute script
F5
OR
(country_of_citizenship NOT IN('Kazakhstan','France') AND EXTRACT(YEAR FROM AGE(date_of_birth))>=19)
)
NOT VALID;
--TASK 8
ALTER TABLE booking ADD COLUMN ticket_discount NUMERIC(9,2);
ALTER TABLE booking ADD CONSTRAINT check_ticket_discount CHECK(
(created_at>'2024-01-01' AND ticket_discount=ROUND(price*0.05,2))
OR
(created_at<'2024-01-01' AND ticket_discount=ROUND(price*0.1,2))
OR
(created_at='2024-01-01')
);
--task 9
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
INSERT INTO booking (booking_id, passenger_id, booking_platform,created_at, update_at, price)
VALUES(501,1,'Price Group','2023-02-02','2024-03-03',10000);
--task 10
ALTER TABLE booking_flight
ALTER COLUMN created_at SET DEFAULT CURRENT_DATE;
INSERT INTO booking_flight(booking_flight_id,booking_id,flight_id,update_at)
VALUES(1001,1,34,CURRENT_DATE);
Data Output
Messages
Notifications
INSERT 0 1
Query returned successfully in 29 msec.
Total rows:
Query complete 00:00:00.029
Ln 79, Col 61

TASK 10


lab4/postgres@PostgreSQL 18*
OR
(country_of_citizenship NOT IN('Kazakhstan','France') AND EXTRACT(YEAR FROM AGE(date_of_birth))>=19)
)
NOT VALID;
--TASK 8
ALTER TABLE booking ADD COLUMN ticket_discount NUMERIC(9,2);
ALTER TABLE booking ADD CONSTRAINT check_ticket_discount CHECK(
(created_at>'2024-01-01' AND ticket_discount=ROUND(price*0.05,2))
OR
(created_at<'2024-01-01' AND ticket_discount=ROUND(price*0.1,2))
OR
(created_at='2024-01-01')
);
--task 9
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
INSERT INTO booking (booking_id, passenger_id, booking_platform,created_at, update_at, price)
VALUES(501,1,'Price Group','2023-02-02','2024-03-03',10000);
--task 10
ALTER TABLE booking_flight
ALTER COLUMN created_at SET DEFAULT CURRENT_DATE;
INSERT INTO booking_flight(booking_flight_id,booking_id,flight_id,update_at)
VALUES(1001,1,34,CURRENT_DATE);
Data Output
Messages
Notifications
INSERT 0 1
Query returned successfully in 42 msec.
Total rows:
Query complete 00:00:00.042
Ln 85, Col 1